Alt-Demazure-atom positivity for the cylindric Hall–Littlewood polynomial $M^{(3)}_{(2,2,0)}$

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Abstract

We prove that the level-3 cylindric Hall–Littlewood polynomial $M^{(3)}_{(2,2,0)}(x_1, x_2, x_3; t)$ admits an explicit expansion in "alt-Demazure atoms" $A^{\mathrm{alt}}_{\gamma}$, indexed by rearrangements γ of the partition $(2, 2, 0)$, with coefficients that are products of Gaussian polynomials $[k]_t = 1+t+\cdots+t^{k-1}$ and that lie in $\mathbb{Z}_{\geq 0}[t]$. Specifically,

$$M^{(3)}_{(2,2,0)}(x;t) \;=\; [2]_t[3]_t\, A^{\mathrm{alt}}_{(2,2,0)}(x;t) \;+\; [2]_t^2\, A^{\mathrm{alt}}_{(2,0,2)}(x;t) \;+\; [2]_t\, A^{\mathrm{alt}}_{(0,2,2)}(x;t).$$

The alt-atom operator θ^{alt}_i is the "$x_i \leftrightarrow tx_{i+1}$-twisted" Mason divided difference; it satisfies a Hecke-type quadratic relation $(\theta^{\mathrm{alt}}_i)^2 = -[2]_t\,\theta^{\mathrm{alt}}_i$ but does *not* satisfy the braid relation, so $A^{\mathrm{alt}}_{\gamma}$ is defined with respect to a fixed reduced-word convention (bubble sort). We record numerical evidence for a broader conjecture: for every partition μ of length $\leq n$ contained in the level-c alcove interior, the classical Hall–Littlewood polynomial $v_\mu(t)P_\mu(x_1, \ldots, x_n; t)$ equals $M^{(c)}_{\mu}$ (interior identity) and expands positively in alt-atoms with Gaussian-product coefficients. This is verified for all μ with $|\mu| \leq 6$ in $n \in \{3, 4\}$.

1 Definitions

Fix $n \geq 2$ and work in the polynomial ring $R = \mathbb{Q}(t)[x_1, \ldots, x_n]$. Let s_i act on R by $(s_if)(x_1, \ldots, x_n) = f(x_1, \ldots, x_{i+1}, x_i, \ldots, x_n)$, swapping x_i and x_{i+1}.

Definition 1 (Alt-atom operator)**.** For $1 \leq i \leq n-1$, define the operator $\theta^{\mathrm{alt}}_i : R \to R$ by

$$\theta^{\mathrm{alt}}_i(f) \;:=\; \frac{x_{i+1} - t\,x_i}{x_i - x_{i+1}}\,\bigl(f - s_if\bigr). \tag{1}$$

Set $\pi^{\mathrm{alt}}_i := \theta^{\mathrm{alt}}_i + \mathrm{id}$.

Lemma 2 (Basic properties)**.** *The operators θ^{alt}_i and π^{alt}_i satisfy the following.*

(a) *θ^{alt}_i maps R to R (the rational singularity at $x_i = x_{i+1}$ is removable: the numerator vanishes on $x_i = x_{i+1}$).*

(b) *At $t = 0$, θ^{alt}_i reduces to the Mason atom operator $\theta^{\mathrm{alt}}_i\big|_{t=0}(f) = \frac{x_{i+1}}{x_i - x_{i+1}}(f - s_if) = \pi_i(f) - f$, where π_i is the classical isobaric Demazure operator.*

(c) *Quadratic (nil-Hecke) relation: $(\theta^{\mathrm{alt}}_i)^2 = -[2]_t\,\theta^{\mathrm{alt}}_i$, i.e. $(\theta^{\mathrm{alt}}_i)^2 + (1+t)\,\theta^{\mathrm{alt}}_i = 0$. Equivalently, π^{alt}_i satisfies $(\pi^{\mathrm{alt}}_i - 1)(\pi^{\mathrm{alt}}_i + t) = 0$, so π^{alt}_i has eigenvalues 1 and $-t$.*

Proof. (a) The polynomial $f - s_i f$ vanishes on $x_i = x_{i+1}$, hence is divisible by $x_i - x_{i+1}$. (b) Direct substitution. (c) Fix f and write $g \coloneqq f - s_i f$; note $s_i g = -g$. Set $r(x) \coloneqq (x_{i+1} - t\,x_i)/(x_i - x_{i+1})$, so $\theta_i^{\mathrm{alt}}(f) = r\,g$. Then

$$\theta_i^{\mathrm{alt}}(r\,g) \;=\; \frac{x_{i+1} - tx_i}{x_i - x_{i+1}}\Big(r\,g - s_i(r\,g)\Big) \;=\; r\,\big(r\,g - (s_i r)(s_i g)\big) \;=\; r\,g\,(r + s_i r).$$

Now $s_i r = (x_i - tx_{i+1})/(x_{i+1} - x_i) = -(x_i - tx_{i+1})/(x_i - x_{i+1})$, so

$$r + s_i r \;=\; \frac{(x_{i+1} - tx_i) - (x_i - tx_{i+1})}{x_i - x_{i+1}} \;=\; \frac{(1+t)(x_{i+1} - x_i)}{x_i - x_{i+1}} \;=\; -(1+t).$$

Therefore $(\theta_i^{\mathrm{alt}})^2(f) = \theta_i^{\mathrm{alt}}(r\,g) = r\,g\,\big(-(1+t)\big) = -(1+t)\,\theta_i^{\mathrm{alt}}(f)$. $\square$

Remark 3 (Braid relations *fail*)**.** Unlike the Mason $\theta_i|_{t=0}$, the operators θ_i^{alt} do *not* satisfy the braid relation $\theta_i^{\mathrm{alt}}\theta_{i+1}^{\mathrm{alt}}\theta_i^{\mathrm{alt}} = \theta_{i+1}^{\mathrm{alt}}\theta_i^{\mathrm{alt}}\theta_{i+1}^{\mathrm{alt}}$ in general. (Direct check on $f = x_1^3x_2^2x_3$ in $n = 4$: both sides differ.) The commutation $\theta_i^{\mathrm{alt}}\theta_j^{\mathrm{alt}} = \theta_j^{\mathrm{alt}}\theta_i^{\mathrm{alt}}$ for $|i-j| \geq 2$ does hold. Consequently, the atom A_γ^{alt} defined below depends on the choice of reduced word, only up to non-adjacent commutations. Throughout this note we fix the *bubble-sort word* convention (Definition **??**).

Definition 4 (Alt-Demazure atom, bubble-sort convention)**.** Let $\gamma = (\gamma_1, \ldots, \gamma_n) \in \mathbb{Z}_{\geq 0}^n$ be a weak composition, and let $\lambda = \mathrm{sort}_\downarrow(\gamma)$ be its decreasing rearrangement (a partition). The *bubble-sort word* $w_{\mathrm{bub}}(\gamma)$ is the sequence of adjacent transpositions produced by the following procedure applied to λ: for each position $p = 1, \ldots, n$, find the leftmost index $i \geq p$ with $\lambda_i = \gamma_p$ (after prior swaps), and bubble that value down to position p using the transpositions $s_i, s_{i-1}, \ldots, s_{p+1}$. This word $(i_1, \ldots, i_\ell)$ has length ℓ equal to the length of the coset $\mathfrak{S}_n/\mathrm{Stab}(\lambda)$ representative for γ. Set

$$A_\gamma^{\mathrm{alt}}(x;t) \;\coloneqq\; \theta_{i_\ell}^{\mathrm{alt}}\theta_{i_{\ell-1}}^{\mathrm{alt}}\cdots\theta_{i_1}^{\mathrm{alt}}\big(x^\lambda\big). \tag{2}$$

When $\gamma = \lambda$ is a partition, $w_{\mathrm{bub}} = \emptyset$ and $A_\lambda^{\mathrm{alt}} = x^\lambda$.

Example 5 (Alt-atoms for $n = 3$, $\lambda = (2,2,0)$)**.** By direct calculation:

$$A_{(2,2,0)}^{\mathrm{alt}}(x;t) = x_1^2x_2^2, \tag{3}$$

$$A_{(2,0,2)}^{\mathrm{alt}}(x;t) = \theta_2^{\mathrm{alt}}(x_1^2x_2^2) \;=\; x_1^2x_3^2 + (1-t)\,x_1^2x_2x_3 - t\,x_1^2x_2^2, \tag{4}$$

$$\begin{aligned} A_{(0,2,2)}^{\mathrm{alt}}(x;t) = \theta_1^{\mathrm{alt}}(\theta_2^{\mathrm{alt}}(x_1^2x_2^2)) \;&=\; x_2^2x_3^2 + (1-t)\,x_1x_2x_3^2 + (1-t)\,x_1x_2^2x_3 \\ &\quad - t\,x_1^2x_3^2 - t(1-t)\,x_1^2x_2x_3. \end{aligned} \tag{5}$$

Verification of Example **??**. For (**??**), set $f = x_1^2x_2^2$; then $s_2 f = x_1^2x_3^2$, so

$$f - s_2 f \;=\; x_1^2(x_2^2 - x_3^2) \;=\; x_1^2(x_2 - x_3)(x_2 + x_3),$$

and dividing by $x_2 - x_3$ gives $\theta_2^{\mathrm{alt}}(f) = (x_3 - tx_2)\cdot x_1^2(x_2 + x_3)$; expanding produces (**??**).

For (**??**), set $f = A_{(2,0,2)}^{\mathrm{alt}}$; then $s_1 f = x_2^2x_3^2 + (1-t)x_1x_2^2x_3 - tx_1^2x_2^2$, and

$$f - s_1 f \;=\; x_3^2(x_1^2 - x_2^2) + (1-t)\,x_1x_2x_3(x_1 - x_2) \;=\; (x_1 - x_2)\big(x_3^2(x_1 + x_2) + (1-t)\,x_1x_2x_3\big).$$

Multiplying by $r_1 = (x_2 - tx_1)/(x_1 - x_2)$ yields $\theta_1^{\mathrm{alt}}(f) = (x_2 - tx_1)\big(x_3^2(x_1 + x_2) + (1-t)\,x_1x_2x_3\big)$, which expands to (**??**). $\square$

Remark 6 (Sanity at $t = 0$)**.** Setting $t = 0$ in (**??**)–(**??**) recovers the Mason atoms $A_{(2,2,0)}|_0 = x_1^2x_2^2$, $A_{(2,0,2)}|_0 = x_1^2x_3^2 + x_1^2x_2x_3$, $A_{(0,2,2)}|_0 = x_2^2x_3^2 + x_1x_2^2x_3 + x_1x_2x_3^2$, whose sum equals the Schur polynomial $s_{(2,2,0)}(x_1, x_2, x_3)$.

2 The cylindric Hall–Littlewood polynomial $M^{(3)}_{(2,2,0)}$

The level-c cylindric Hall–Littlewood polynomials $M^{(c)}_\mu$ are defined by iterating the van Diejen–Emsiz–Zurrián (vDEZ) Corollary 4.2 Pieri rule from $M^{(c)}_\emptyset = 1$ (see [?]). For our purposes we adopt the following formula, verified against vDEZ in [?, 07-21 ter memo]:

Proposition 7 (Explicit $M^{(3)}_{(2,2,0)}$). *Set $m_{(2,2,0)} = x_1^2x_2^2 + x_1^2x_3^2 + x_2^2x_3^2$ (monomial symmetric) and $e_3e_1 = x_1x_2x_3(x_1 + x_2 + x_3)$. Then*

$$M^{(3)}_{(2,2,0)}(x_1, x_2, x_3; t) \;=\; (1 + t)\Big(m_{(2,2,0)} + (1 - t)\, e_3e_1\Big). \tag{6}$$

Moreover, $M^{(3)}_{(2,2,0)} = v_{(2,2,0)}(t)\, P_{(2,2,0)}(x_1, x_2, x_3; t)$, the classical Hall–Littlewood polynomial with Macdonald's normalization $v_{(2,2,0)}(t) = [2]_t$ (since the multiplicity of 2 in $(2, 2, 0)$ is 2).

The identity $M^{(c)}_\mu = v_\mu P_\mu$ holds for all μ in the interior of the level-c alcove, i.e. $\mu_1 - \mu_n < c$. For $\mu = (2, 2, 0)$ and $c = 3$ we have $\mu_1 - \mu_n = 2 < 3$, so $(2, 2, 0)$ is interior; the identity is verified numerically in every case we tested (see §**??**). We treat (**??**) as our working definition of the target.

3 Main theorem

Theorem 8 (Alt-atom expansion of $M^{(3)}_{(2,2,0)}$). *In the polynomial ring $\mathbb{Q}(t)[x_1, x_2, x_3]$,*

$$M^{(3)}_{(2,2,0)}(x_1, x_2, x_3; t) \;=\; [2]_t[3]_t\, A^{\mathrm{alt}}_{(2,2,0)} \;+\; [2]_t^2\, A^{\mathrm{alt}}_{(2,0,2)} \;+\; [2]_t\, A^{\mathrm{alt}}_{(0,2,2)}. \tag{7}$$

All three coefficients are products of Gaussian polynomials, and lie in $\mathbb{Z}_{\geq 0}[t]$.

Proof. Both sides of (**??**) are polynomials in $\mathbb{Q}(t)[x_1, x_2, x_3]$, homogeneous of degree 4. It suffices to verify equality of coefficients on the monomial basis $\{x_1^a x_2^b x_3^c\}$. The left-hand side, from Proposition **??**, expands as

$$\begin{aligned}(1 + t)\big(x_1^2x_2^2 + x_1^2x_3^2 + x_2^2x_3^2\big) + (1 + t)(1 - t)\big(x_1^2x_2x_3 + x_1x_2^2x_3 + x_1x_2x_3^2\big)\\ = (1 + t)\, m_{(2,2,0)} + (1 - t^2)\, m_{(2,1,1)}.\end{aligned}$$

The right-hand side, using (**??**)–(**??**), is

$$\begin{aligned}[2]_t[3]_t\, A^{\mathrm{alt}}_{(2,2,0)} &= (1 + t)(1 + t + t^2)\, x_1^2x_2^2,\\ [2]_t^2\, A^{\mathrm{alt}}_{(2,0,2)} &= (1 + t)^2\big(x_1^2x_3^2 + (1 - t)\, x_1^2x_2x_3 - t\, x_1^2x_2^2\big),\\ [2]_t\, A^{\mathrm{alt}}_{(0,2,2)} &= (1 + t)\big(x_2^2x_3^2 + (1 - t)\, x_1x_2x_3^2 + (1 - t)\, x_1x_2^2x_3 - t\, x_1^2x_3^2 - t(1 - t)\, x_1^2x_2x_3\big).\end{aligned}$$

We tabulate the coefficient of each monomial:

$x_1^2x_2^2$: $(1 + t)(1 + t + t^2) - t(1 + t)^2 = (1 + t)\big[(1 + t + t^2) - t(1 + t)\big] = (1 + t) \cdot 1 = 1 + t.$
LHS: $1 + t$. *Match.*

$x_1^2x_3^2$: $(1 + t)^2 - t(1 + t) = (1 + t)\big[(1 + t) - t\big] = (1 + t).$
LHS: $1 + t$. *Match.*

$x_2^2x_3^2$: $(1 + t) \cdot 1 = 1 + t.$
LHS: $1 + t$. *Match.*

$x_1^2x_2x_3$**:** $(1+t)^2(1-t) - t(1-t)(1+t) = (1+t)(1-t)\big[(1+t)-t\big] = (1+t)(1-t) = 1-t^2$.
LHS: $1-t^2$. *Match.*

$x_1x_2^2x_3$**:** $(1+t)(1-t) = 1-t^2$.
LHS: $1-t^2$. *Match.*

$x_1x_2x_3^2$**:** $(1+t)(1-t) = 1-t^2$.
LHS: $1-t^2$. *Match.*

There are no other monomials of degree 4 supported by either side. Hence both sides of (**??**) are equal as polynomials in $\mathbb{Q}(t)[x_1,x_2,x_3]$.

Each coefficient $[2]_t[3]_t = 1+2t+2t^2+t^3$, $[2]_t^2 = 1+2t+t^2$, and $[2]_t = 1+t$ is a product of Gaussians and lies in $\mathbb{Z}_{\geq 0}[t]$. $\square$

Remark 9 (Cross-checks)**.** The identity (**??**) was independently verified by symbolic computation in Python/SymPy (`atoms_engine.py`, `atoms_hl_alternatives.py` at `~/projects/probes/2026-07-22-atoms-pro`
At the specializations $t=0$ and $t=1$ both sides give the correct classical limits: at $t=0$, the sum $A^{\text{alt}}_{(2,2,0)} + A^{\text{alt}}_{(2,0,2)} + A^{\text{alt}}_{(0,2,2)} = s_{(2,2,0)}(x_1,x_2,x_3)$ recovers the Schur polynomial (Mason atom decomposition of a Schur function); at $t=1$, both sides equal $2\,m_{(2,2,0)}(x_1,x_2,x_3)$.

4 Extensions and the general conjecture

The identity (**??**) is the simplest interior case of a much broader pattern. Setting $M_\mu^{(c)} = v_\mu(t)\,P_\mu(x_1,\dots,x_n;t)$ (the classical Hall–Littlewood polynomial with Macdonald's v_μ-normalizer $v_\mu(t) = \prod_{i\geq 0}[m_i(\mu)]_t!$), the following holds numerically for every case tested.

Theorem 10 (Verified data)**.** *For (n,μ) in each of the following cases, the classical Hall–Littlewood expression $v_\mu(t)P_\mu(x_1,\dots,x_n;t)$ admits a t-positive alt-atom expansion*

$$v_\mu(t)\,P_\mu(x;t) \;=\; \sum_{\gamma\in\mathfrak{S}_n\cdot\mu} c_\gamma(t)\,A^{\text{alt}}_\gamma(x;t), \qquad c_\gamma(t)\in\mathbb{Z}_{\geq 0}[t],$$

with each $c_\gamma(t)$ a product of Gaussian polynomials $[k]_t$:

- $n=3$, $\mu\in\{(2,1,0),\ (2,2,0),\ (2,1,1),\ (3,0,0),\ (2,2,1),\ (2,2,2),\ (3,1,1),\ (3,2,0),\ (3,2,1)\}$;
- $n=4$, $\mu\in\{(2,1,0,0),\ (2,2,0,0),\ (2,1,1,0),\ (2,2,1,0),\ (3,0,0,0),\ (3,2,1,0),\ (2,2,2,0)\}$.

Each identity is a finite polynomial identity, verified by symbolic computation.

For example, at $n=3$ and $\mu=(2,1,0)$ (all parts distinct, six rearrangements):

$$\begin{aligned} P_{(2,1,0)}(x_1,x_2,x_3;t) \;=\; & [2]_t[3]_t\,A^{\text{alt}}_{(2,1,0)} + [3]_t\,A^{\text{alt}}_{(2,0,1)} + [2]_t^2\,A^{\text{alt}}_{(1,2,0)} \\ & + [2]_t\,A^{\text{alt}}_{(1,0,2)} + [2]_t\,A^{\text{alt}}_{(0,2,1)} + A^{\text{alt}}_{(0,1,2)}. \end{aligned}$$

Note $v_{(2,1,0)} = 1$ (all multiplicities are 1), so this is P_μ directly.

Conjecture 11 (Interior atoms positivity)**.** Fix $n\geq 2$ and a level $c\geq 1$. For every partition $\mu = (\mu_1\geq\cdots\geq\mu_n\geq 0)$ with $\mu_1-\mu_n<c$ (*alcove interior*), one has $M_\mu^{(c)} = v_\mu(t)P_\mu(x_1,\dots,x_n;t)$, and

$$M_\mu^{(c)}(x;t) \;=\; \sum_{\gamma\in\mathfrak{S}_n\cdot\mu} c_\gamma(t)\,A^{\text{alt}}_\gamma(x;t),$$

where each $c_\gamma(t)\in\mathbb{Z}_{\geq 0}[t]$ is a product of Gaussian polynomials $[k]_t$ for $k=1,2,\dots,n$.

4.1 The boundary case $\mu = (3,1,0)$

For μ on the alcove boundary ($\mu_1 - \mu_n = c$), the cylindric structure introduces a correction. We verified numerically:

$$M^{(3)}_{(3,1,0)}(x_1,x_2,x_3;t) \;=\; \frac{[2]_t}{[3]_t}\, P_{(3,1,0)}(x_1,x_2,x_3;t),$$

which is *not* $v_\mu(t)P_\mu = P_{(3,1,0)}$ (since $v_{(3,1,0)} = 1$). The boundary factor $[2]_t/[3]_t$ absorbs the cylindric correction. Multiplying by $[3]_t = 1+t+t^2$ restores integer-Gaussian atom positivity:

$$\begin{aligned}[3]_t\, M^{(3)}_{(3,1,0)} \;&=\; [2]_t[3]_t\, A^{\mathrm{alt}}_{(3,1,0)} + [2]_t[3]_t\, A^{\mathrm{alt}}_{(3,0,1)} + [2]_t^3\, A^{\mathrm{alt}}_{(1,3,0)} \\ &\quad + [2]_t^2\, A^{\mathrm{alt}}_{(1,0,3)} + [2]_t^2\, A^{\mathrm{alt}}_{(0,3,1)} + [2]_t\, A^{\mathrm{alt}}_{(0,1,3)}.\end{aligned}$$

All coefficients are Gaussian products.

Conjecture 12 (Boundary atoms positivity)**.** For μ with $\mu_1 - \mu_n = c$ (alcove boundary), there is an explicit Gaussian denominator $D_\mu(t) \in \mathbb{Z}_{\geq 0}[t]$ (a product of $[k]_t$'s) such that $D_\mu(t)\, M^{(c)}_\mu(x;t)$ is a t-positive Gaussian-coefficient linear combination of alt-atoms. For $\mu = (3,1,0)$, $D_\mu(t) = [3]_t$.

5 Structural observations

5.1 Where the Gaussian factors come from

The alt-atom quadratic $(\theta^{\mathrm{alt}}_i)^2 = -[2]_t\theta^{\mathrm{alt}}_i$ (Lemma **??**(c)) is the principal source of $[2]_t$ factors: whenever θ^{alt}_i acts on a Bruhat cover cell where its action "turns around" (via $-[2]_t$), a $[2]_t$ factor emerges. The $[k]_t$ factors for $k \geq 3$ arise from compositions across multiple Bruhat descents.

In the dominant case $\gamma = \lambda$ (identity coset representative), the coefficient $c_\lambda(t)$ equals the Gaussian factorial $[n]_t! = [n]_t[n-1]_t \cdots [1]_t$ in every observed case. This matches the Cherednik–Ram identity $\sum_{w\in\mathfrak{S}_n} T_w(x^\mu) = v_\mu(t)P_\mu(x;t)$ under $A^{\mathrm{alt}}_\lambda = x^\lambda$: the total Hecke weight of $\mathfrak{S}_n$ is $[n]_t!$.

5.2 Relation to Blasiak–Haglund–Pun–Sergel

Blasiak, Haglund, Pun, and Sergel conjectured [**?**] that certain nonsymmetric flagged LLT polynomials expand positively in Demazure atoms. Our cylindric-HL polynomial $M^{(c)}_\mu$ is a symmetric-function shadow of a specific class of 2-ribbon LLTs (via the ribbon-Fock correspondence). Theorem **??** and the extended evidence in Theorem **??** provide first evidence in the *cylindric HL* direction of their broader positivity program.

5.3 Relation to Assaf–González

At $t = 0$, our identity $M^{(c)}_\mu|_{t=0} = \sum_\gamma A^{\mathrm{alt}}_\gamma|_{t=0}$ (unit coefficients) is the classical Demazure-union statement $s_\mu = \sum_\gamma A_\gamma$ [**?**]. At general t, the $[k]_t$-weighted sum can be viewed as a graded refinement of this Demazure union, matching in spirit the edge-local criterion of Assaf–González [**?**]. Making this connection rigorous would identify the Gaussian weight $[k]_t$ with the graded character of a specific rank-k Demazure subquotient.

6 Gaps

- **Braid non-invariance.** The atom A_γ^{alt} depends on the reduced-word convention (Remark **??**); we fix bubble sort. A "correct" atom convention that satisfies braid *and* yields positivity is not yet identified.

- **Uniform proof.** Conjectures **??** and **??** are verified numerically but not proved in general. A natural approach is by induction on the vDEZ Corollary 4.2 Pieri rule, but this requires proving that $e_r \cdot A_\gamma^{\mathrm{alt}}$ expands t-positively in atoms (a t-graded analogue of the classical Pieri rule for Demazure atoms, established for keys by Reiner–Shimozono [**?**] and by Assaf [**?**]).

- **Closed formula for $c_\gamma(t)$.** We do not have a combinatorial formula determining $c_\gamma(t)$ from γ. Numerically, c_γ depends on more than just Bruhat length: it varies with the specific descent path from $\lambda = \mathrm{sort}_\downarrow(\gamma)$ to γ. For example in $n = 3$, $\mu = (2, 1, 0)$: both $\gamma = (2, 0, 1)$ and $\gamma = (1, 2, 0)$ have Bruhat length 1, but $c_{(2,0,1)} = [3]_t$ while $c_{(1,2,0)} = [2]_t^2$.

- **Identification of θ_i^{alt}.** The operator θ_i^{alt} is a specific t-deformation of Mason's atom operator, but has not been matched to any named operator in Assaf–Schilling [**?**] or in Blasiak et al. definitions. It is *not* the Cherednik–Lusztig T_i (which satisfies $T_i^2 = (t-1)T_i + t$), nor the Ram–Ion isobaric π_i^t used in the "standard" HL atom convention (which gives non-positive expansions with poles at $t = 1$; see `atoms_hl_alternatives.py`).

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